Solutions, Eighth Annual ECC Undergraduate Mathematics Competition, April 9, 2005

1. Sum of digits.

The sum is 18045. We may write $n = 10^{2005} - 1$, so

$$n^2 = 10^{4010} - 2 \cdot 10^{2005} + 1 = 10^{2005}(10^{2005} - 2) + 1.$$

Now, $10^{2005}-2$ in decimal form is $999\ldots998$, with a total of 2005 digits. Multiplying by 10^{2005} puts 2005 zeros following the 8, and adding 1 changes the final 0 to 1. Thus, n^2 begins with 2004 9s followed by one 8, 2004 0s and a 1, and the sum is $2004 \cdot 9 + 8 + 1 = 2005 \cdot 9 = 18045$.

2. Limit of a sequence.

The limit is 1/2. We have

$$n(\sqrt{n^2+1} - n) = \frac{n}{\sqrt{n^2+1} + n} = \frac{1}{\sqrt{1 + \frac{1}{n^2}} + 1} \to \frac{1}{\sqrt{1+0}+1} = \frac{1}{2}$$

as $n \to \infty$.

3. Find a formula.

The sum is $n(n^2+1)/2$, as we show: The total number of integers in the first n subsets is

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

The n-th subset has n integers, the last of which is $l = n(n+1)/2$. Then the first is $a = n(n+1)/2 - (n-1)$, and the sum is

$$g(n) = \frac{n(a+l)}{2} = \frac{n[n(n+1) - n + 1]}{2} = \frac{n(n^2+1)}{2}.$$

4. Integer solutions.

Integer solutions exist for $m = -2$ and for $m = 6$, and for these values only. Suppose integer solutions exist. Eliminating y between the two equations we find that

$$(13 + 11m)x = 711 = 9 \cdot 79.$$

For x to be an integer we need

$$13 + 11m \in \{\pm 1, \pm 3, \pm 9, \pm 79, \pm 237, \pm 711\}.$$

Now, -13 ± 1, -13 ± 3, $-13 + 9$, $-13 - 79$, -13 ± 237 and -13 ± 711 are not multiples of 11, but $-13 - 9 = -22 = 11(-2)$ and $-13 + 79 = 66 = 11 \cdot 6$. Thus -2 and 6 are the only possible values of m. With $m = -2$ we have the solution $x = -79$, $y = 157$, and with $m = 6$ the solution $x = 9$, $y = 53$.

5. Integral equation.

The unique solution is $x = 7604$. For $10^k \le t < 10^{k+1}$, $\lfloor \log_{10} t \rfloor = k$, so

$$\begin{aligned}\int_1^{10^3} \frac{dt}{1 + \lfloor \log_{10} t \rfloor} &= \int_1^{10} dt + \int_{10}^{10^2} \frac{dt}{2} + \int_{10^2}^{10^3} \frac{dt}{3} \\ &= 9 + \frac{90}{2} + \frac{900}{3} = 354.\end{aligned}$$

For $10^3 < x < 10^4$,

$$\int_{10^3}^{x} \frac{dt}{1 + \lfloor \log_{10} t \rfloor} = \frac{x - 1000}{4},$$

so we need $\frac{x-1000}{4} = 2005 - 354 = 1651$, and the unique solution is

$$x = (4)(1651) + 1000 = 7604.$$

6. It's an integer.

We prove by induction that $r^n + s^n$ is an integer for every positive integer n. From

$$x^2 - 9x + 1 = (x - r)(x - s) = x^2 - (r + s)x + rs$$

we see that $r + s = 9$. From $r^2 = 9r - 1$ and $s^2 = 9s - 1$ we have

$$r^2 + s^2 = 9(r + s) - 2,$$

which is an integer. Suppose that $r^k + s^k$ is an integer for $1 \le k \le n$. Then

$$r^{k+1} = 9r^k - r^{k-1} \quad \text{and} \quad s^{k+1} = 9s^k - s^{k-1},$$

so

$$r^{k+1} + s^{k+1} = 9(r^k + s^k) - (r^{k-1} + s^{k-1})$$

is an integer. By induction, $r^n + s^n$ is an integer for all $n \ge 1$. ∎

7. An inequality.

Rewrite the expression as follows:

$$\begin{aligned}\frac{(a+b)^3}{a^2b} &= \frac{a^3 + 3a^2b + 3ab^2 + b^3}{a^2b} \\ &= \frac{a}{b} + 3 + 3\frac{b}{a} + \frac{b^2}{a^2} \\ &= u^2 + 3u + 3 + \frac{1}{u} = f(u), \text{ say}\end{aligned}$$

where $u = \frac{b}{a} > 0$. Now

$$\begin{aligned}f'(u) &= 2u + 3 - \frac{1}{u^2} \\ &= \frac{1}{u^2}(2u^3 + 3u - 1) \\ &= \frac{1}{u^2}(u+1)^2(2u-1).\end{aligned}$$

From this we see that $f'(u) < 0$ in $(0, \frac{1}{2})$, $f'(\frac{1}{2}) = 0$, and $f'(u) > 0$ for $u > \frac{1}{2}$, so $f(u)$ has a global minimum for positive u at $u = \frac{1}{2}$. Since $f(\frac{1}{2}) = \frac{27}{4}$, it follows that $f(u) \ge \frac{27}{4}$ for all $u > 0$ and $f(u) > \frac{27}{4}$ when $u > 0$ but $u \ne \frac{1}{2}$. Thus

$$\frac{(a+b)^3}{a^2b} \ge \frac{27}{4},$$

with equality if and only if $u = \frac{b}{a} = \frac{1}{2}$.

8. Minimum area ratio.

We show that the minimum value of r is 2. Independently of the rectangle S, the rectangle T has its smallest area when its sides are on the lines $x = \pm a$ and $y = \pm b$, in which case its area is $4ab$. To complete the proof it suffices to show that the maximum possible area of S is $2ab$. It is clear that when S has maximum area, its vertices are on the ellipse. Let (x, y) be the first quadrant vertex on the ellipse. Then Area(S) is $4xy$, and (x, y) satisfies

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

When $4xy$ is a maximum, so is $\frac{x^2y^2}{b^2}$, and

$$\frac{x^2y^2}{b^2} = x^2\Big(1 - \frac{x^2}{a^2}\Big) = f(x), \text{ say;} \quad 0 \le x \le a.$$

We see that $f(0) = f(a) = 0$, so the maximum occurs at an interior point of the interval, and $f'(x)$ must vanish at such a point. Well,

$$f'(x) = 2x - \frac{4x^3}{a^2} = 2x\Big(1 - \frac{2x^2}{a^2}\Big),$$

which vanishes in the interior if and only if $x^2 = a^2/2$. Then $y^2 = b^2/2$, and $x^2y^2 = a^2b^2/4$; $xy = ab/2$, and Area$(S) = 4xy = 2ab$. ∎

SECOND SOLUTION (without calculus)

By the arithmetic-geometric mean inequality,

$$\frac{xy}{ab} \le \frac{1}{2}\Big(\frac{x^2}{a^2} + \frac{y^2}{b^2}\Big) = \frac{1}{2}. \tag{1}$$

When $\frac{x^2}{a^2} = \frac{y^2}{b^2}$ we have equality in (1), so the maximum value of xy is $ab/2$. This makes the maximum value of Area$(S) = 2ab$ as before.

9. Probability it's divisible by 10.

The answer is

$$1 - \left(\frac{8}{9}\right)^n - \left(\frac{5}{9}\right)^n + \left(\frac{4}{9}\right)^n.$$

The product is a multiple of 10 if and only if at least one 5 and at least one even integer have been generated. If A is the event that a 5 has been generated and B the event that an even number has been generated, it is $\Pr(A \cap B)$ that we seek. Now

$$\begin{aligned}\Pr((A \cap B)') &= \Pr(A' \cup B') \\ &= \Pr(A') + \Pr(B') - \Pr(A' \cap B').\end{aligned}$$

$A' \cap B'$ is the event that neither a 5 nor an even integer has been generated, and $\Pr(A' \cap B') = (\frac{4}{9})^n$. It is easy to see that $\Pr(A') = (\frac{8}{9})^n$ and $\Pr(B') = (\frac{5}{9})^n$. Thus

$$\Pr((A \cap B)') = \left(\frac{8}{9}\right)^n + \left(\frac{5}{9}\right)^n - \left(\frac{4}{9}\right)^n,$$

and

$$\Pr(A \cap B) = 1 - \Pr((A \cap B)') = 1 - \left(\frac{8}{9}\right)^n - \left(\frac{5}{9}\right)^n + \left(\frac{4}{9}\right)^n.$$

10. Radical equation.

The unique root is $x = 2$. The function $f(x) = \sqrt{2+x}$ is being iterated. This function has $x = 2$ as a fixed point: $f(2) = 2$. We will show that (i) if $x > 2$, then $2 < f(x) < x$, and (ii) if $0 < x < 2$, then $x < f(x) < 2$. From these two facts it follows that if $x \neq 2$ then applying f repeatedly to x will never return to the original value x.

(i) Suppose that $x > 2$. Then

$$f(x) = \sqrt{2+x} > \sqrt{2+2} = 2.$$

Also,

$$(f(x))^2 - x^2 = x + 2 - x^2 = -(x-2)(x+1) < 0,$$

so $(f(x))^2 < x^2$ and $f(x) < x$. Thus $2 < f(x) < x$.

(ii) Suppose that $0 < x < 2$. Then $f(x) = \sqrt{2+x} < 2$, and

$$(f(x))^2 - x^2 = -(x-2)(x+1) > 0,$$

so $x < f(x) < 2$, and the proof is complete.